

ERP System — Complete Frontend Architecture

React 18 + Vite + TailwindCSS + Zustand + TanStack Query + Socket.io Client

Role: React Tech Lead Design Document

Phase: Pre-code Architecture (No code generated yet)

Modules covered: All 17 ERP modules | 4 Roles: Admin, Sales Manager, Inventory Manager, Dispatch Worker

1. COMPLETE FOLDER STRUCTURE

```
erp-frontend/
├── public/
│   ├── favicon.ico
│   ├── logo.svg
│   └── manifest.json
│
├── src/
│   ├── main.jsx                # App entry point
│   ├── App.jsx                 # Root component, provider tree
│   │
│   ├── assets/                 # Static assets
│   │   ├── images/
│   │   ├── icons/
│   │   └── fonts/
│   │
│   ├── config/                 # App-level configuration
│   │   ├── app.config.js
│   │   ├── routes.config.js
│   │   ├── permissions.config.js
│   │   └── socket.config.js
│   │
│   ├── constants/              # App-wide constants
│   │   └── roles.js
```

```

|   |   |── permissions.js
|   |   |── orderStatus.js
|   |   |── dispatchStatus.js
|   |   |── notificationTypes.js
|   |   |── stockTypes.js
|   |   └── index.js
|
|   |
|   |── theme/                                # Design system & theming
|   |   |── colors.js
|   |   |── typography.js
|   |   |── spacing.js
|   |   |── breakpoints.js
|   |   |── shadows.js
|   |   |── tokens.js
|   |   └── tailwind.preset.js
|
|   |
|   |── router/                              # Routing architecture
|   |   |── index.jsx                        # Root router definition
|   |   |── ProtectedRoute.jsx              # Auth + role guard
|   |   |── PermissionRoute.jsx             # Permission-level guard
|   |   |── RoleRouter.jsx                  # Role-to-layout dispatcher
|   |   └── routes/
|   |       |── authRoutes.jsx
|   |       |── adminRoutes.jsx
|   |       |── salesRoutes.jsx
|   |       |── inventoryRoutes.jsx
|   |       └── dispatchRoutes.jsx
|
|   |
|   |── layouts/                             # Shell layouts per role
|   |   |── AuthLayout/
|   |   |   |── index.jsx
|   |   |   └── AuthLayout.module.css
|   |   |── AdminLayout/
|   |   |   |── index.jsx
|   |   |   |── AdminSidebar.jsx
|   |   |   |── AdminTopbar.jsx
|   |   |   └── AdminFooter.jsx
|   |   |── SalesLayout/
|   |   |   |── index.jsx
|   |   |   |── SalesSidebar.jsx
|   |   |   └── SalesTopbar.jsx

```

```

|   |   |   └── InventoryLayout/
|   |   |       ├── index.jsx
|   |   |       ├── InventorySidebar.jsx
|   |   |       └── InventoryTopbar.jsx
|   |   └── DispatchLayout/
|   |       ├── index.jsx
|   |       ├── DispatchSidebar.jsx
|   |       └── DispatchTopbar.jsx
|   └── shared/
|       ├── Sidebar.jsx           # Base sidebar (role fills it)
|       ├── Topbar.jsx           # Base topbar (role fills it)
|       ├── NotificationBell.jsx
|       ├── UserMenu.jsx
|       └── BreadcrumbBar.jsx
|
|   └── features/                  # Feature-based modules (core)
|       |
|       ├── auth/
|           ├── pages/
|               ├── LoginPage.jsx
|               └── ForgotPasswordPage.jsx
|           ├── components/
|               ├── LoginForm.jsx
|               └── ForgotPasswordForm.jsx
|           ├── hooks/
|               └── useAuth.js
|           ├── api/
|               └── authApi.js
|           └── store/
|               └── authStore.js    # Zustand auth slice
|
|       ├── dashboard/
|           ├── pages/
|               ├── AdminDashboard.jsx
|               ├── SalesDashboard.jsx
|               ├── InventoryDashboard.jsx
|               └── DispatchDashboard.jsx
|           ├── components/
|           ├── widgets/
|               ├── StatCard.jsx
|               └── RevenueChart.jsx

```

```
| | | | | |—— OrderStatusPie.jsx
| | | | | |—— StockLevelBar.jsx
| | | | | |—— CreditUtilizationWidget.jsx
| | | | | |—— DispatchQueueWidget.jsx
| | | | | |—— ReorderAlertWidget.jsx
| | | | | |—— SmartSuggestionWidget.jsx
| | | | |—— DashboardGrid.jsx
| | | |—— hooks/
| | | |—— useDashboardStats.js
| | |—— api/
| | |—— dashboardApi.js
| | |
| |—— users/
| | |—— pages/
| | | |—— UserListPage.jsx
| | | |—— UserCreatePage.jsx
| | | |—— UserEditPage.jsx
| | | |—— components/
| | | |—— UserTable.jsx
| | | |—— UserForm.jsx
| | | |—— UserStatusToggle.jsx
| | | |—— hooks/
| | | |—— useUsers.js
| | |—— api/
| | |—— usersApi.js
| | |
| |—— roles/
| | |—— pages/
| | | |—— RolePermissionsPage.jsx
| | | |—— components/
| | | |—— PermissionMatrix.jsx
| | | |—— RoleCard.jsx
| | | |—— hooks/
| | | |—— useRoles.js
| | |—— api/
| | |—— rolesApi.js
| | |
| |—— customers/
| | |—— pages/
| | | |—— CustomerListPage.jsx
| | | |—— CustomerCreatePage.jsx
```

```
| | | | | CustomerEditPage.jsx
| | | | | CustomerDetailPage.jsx
| | | | | components/
| | | | | CustomerTable.jsx
| | | | | CustomerForm.jsx
| | | | | CustomerCreditPanel.jsx
| | | | | CustomerOrderHistory.jsx
| | | | | hooks/
| | | | | useCustomers.js
| | | | | api/
| | | | | customersApi.js
| | |
| | | vendors/
| | | | | pages/
| | | | | VendorListPage.jsx
| | | | | VendorCreatePage.jsx
| | | | | VendorDetailPage.jsx
| | | | | components/
| | | | | VendorTable.jsx
| | | | | VendorForm.jsx
| | | | | hooks/
| | | | | useVendors.js
| | | | | api/
| | | | | vendorsApi.js
| | |
| | | products/
| | | | | pages/
| | | | | ProductListPage.jsx
| | | | | ProductCreatePage.jsx
| | | | | ProductEditPage.jsx
| | | | | ProductDetailPage.jsx
| | | | | components/
| | | | | ProductTable.jsx
| | | | | ProductForm.jsx
| | | | | CategoryTree.jsx
| | | | | ProductStockBadge.jsx
| | | | | hooks/
| | | | | useProducts.js
| | | | | api/
| | | | | productsApi.js
| | |
```

```
| | |—— inventory/
| | | |—— pages/
| | | | |—— StockListPage.jsx
| | | | |—— StockTransactionPage.jsx
| | | | |—— StockAdjustmentPage.jsx
| | | |—— components/
| | | | |—— Stock1Table.jsx
| | | | |—— Stock2Table.jsx
| | | | |—— StockComparisonCard.jsx
| | | | |—— TransactionLedger.jsx
| | | | |—— StockAdjustmentForm.jsx
| | | |—— hooks/
| | | | |—— useInventory.js
| | | |—— api/
| | | | |—— inventoryApi.js
| | |
| | |—— orders/
| | | |—— pages/
| | | | |—— OrderListPage.jsx
| | | | |—— OrderCreatePage.jsx
| | | | |—— OrderDetailPage.jsx
| | | | |—— OrderEditPage.jsx
| | | |—— components/
| | | | |—— OrderTable.jsx
| | | | |—— OrderForm.jsx
| | | | |—— OrderLineItemsEditor.jsx
| | | | |—— OrderStatusBadge.jsx
| | | | |—— OrderStatusStepper.jsx
| | | | |—— OrderActionBar.jsx
| | | | |—— CreditHoldBanner.jsx
| | | | |—— OrderSummaryPanel.jsx
| | | |—— hooks/
| | | | |—— useOrders.js
| | | | |—— useOrderRealtime.js # Socket-powered live updates
| | | |—— api/
| | | | |—— ordersApi.js
| | |
| | |—— dispatch/
| | | |—— pages/
| | | | |—— DispatchQueuePage.jsx
| | | | |—— DispatchDetailPage.jsx
```

```
| | | | └── DispatchHistoryPage.jsx
| | | | └── components/
| | | | | └── DispatchQueue.jsx
| | | | | └── DispatchCard.jsx
| | | | | └── DispatchStatusStepper.jsx
| | | | | └── PicklistPanel.jsx
| | | | | └── DeliveryConfirmForm.jsx
| | | | └── hooks/
| | | | | └── useDispatch.js
| | | | | └── useDispatchRealtime.js
| | | | └── api/
| | | | | └── dispatchApi.js
| | |
| | └── payments/
| | | └── pages/
| | | | └── PaymentListPage.jsx
| | | | └── RecordPaymentPage.jsx
| | | | └── components/
| | | | | └── PaymentTable.jsx
| | | | | └── RecordPaymentForm.jsx
| | | | | └── PaymentReceiptCard.jsx
| | | | └── hooks/
| | | | | └── usePayments.js
| | | | └── api/
| | | | | └── paymentsApi.js
| | |
| | └── credit/
| | | └── pages/
| | | | └── CreditManagementPage.jsx
| | | | └── components/
| | | | | └── CreditLimitTable.jsx
| | | | | └── CreditLimitForm.jsx
| | | | | └── CreditHistoryDrawer.jsx
| | | | | └── CreditRequestBanner.jsx
| | | | └── hooks/
| | | | | └── useCredit.js
| | | | └── api/
| | | | | └── creditApi.js
| | |
| | └── reorder/
| | | └── pages/
```

```
| | | | └── ReorderPage.jsx
| | | └── components/
| | | | └── ReorderTable.jsx
| | | | └── ReorderActionButtons.jsx
| | | | └── ReorderThresholdForm.jsx
| | | └── hooks/
| | | | └── useReorder.js
| | | └── api/
| | | | └── reorderApi.js
| | |
| | └── suggestions/
| | | └── pages/
| | | | └── SuggestionsPage.jsx
| | | └── components/
| | | | └── SuggestionFeed.jsx
| | | | └── SuggestionCard.jsx
| | | └── hooks/
| | | | └── useSuggestions.js
| | | └── api/
| | | | └── suggestionsApi.js
| | |
| | └── notifications/
| | | └── pages/
| | | | └── NotificationsPage.jsx
| | | └── components/
| | | | └── NotificationList.jsx
| | | | └── NotificationItem.jsx
| | | | └── NotificationDropdown.jsx
| | | └── hooks/
| | | | └── useNotifications.js
| | | | └── useNotificationSocket.js
| | | └── api/
| | | | └── notificationsApi.js
| | | └── store/
| | | | └── notificationStore.js # Zustand notification slice
| | |
| | └── audit/
| | | └── pages/
| | | | └── AuditLogPage.jsx
| | | └── components/
| | | | └── AuditLogTable.jsx
```



```
| | | | └── AuditDetailDrawer.jsx
| | | | └── hooks/
| | | |   └── useAuditLogs.js
| | | | └── api/
| | | |   └── auditApi.js
| | |
| | └── excel/
| |   └── pages/
| |     └── ExcelImportExportPage.jsx
| |   └── components/
| |     └── ImportUploadZone.jsx
| |     └── ImportProgressTracker.jsx
| |     └── ImportErrorReport.jsx
| |     └── ExportButtons.jsx
| |   └── hooks/
| |     └── useExcelImport.js
| |   └── api/
| |     └── excelApi.js
| |
| └── reports/
|   └── pages/
|     └── ReportsPage.jsx
|   └── components/
|     └── ReportSelector.jsx
|     └── PDFDownloadButton.jsx
|
```